

An LLM-Augmented R/Shiny Platform for Quantitative Systems Pharmacology

2026 Annual Conference of the Korean Society for Mathematical
Biology (KSMB) · Special Session B1 — Mathematical Modeling in New
Drug Development & Clinical Pharmacotherapy

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Roadmap

KSMB 2026 · Special Session B1

Thesis. *Quantitative Systems Pharmacology (QSP) turns drug action into large systems of nonlinear ODEs. Building them is slow and rare. We let an autonomous LLM agent build them —*
191 disease models and counting.

1. **The problem** — why drug development fails, and why statistics aren't enough
2. **QSP and its mathematics** — $dx/dt = f(x, \theta, u)$
3. **The engine** — the Claude Code Routine, an autonomous model-builder
4. **Why an LLM** — tireless, broad, consistent
5. **Scale & breadth** — a living library across ~15 therapeutic areas
6. **Deep dives** — IgA nephropathy · sickle cell · multiple myeloma
7. **Open mathematical problems** — where *this room* comes in

Drug development is staggeringly expensive — and mostly fails

- ~\$1-2.6 billion and 10-15 years per approved drug
 - ~90% of candidates entering clinical trials never reach patients
 - Failures cluster at two points:
 - **Early** — the target was never mechanistically validated
 - **Late** — unexpected efficacy / safety surprises in pivotal trials
- ~90% clinical-phase attrition
~\$2.6B cost per approved drug

We need models that **explain and extrapolate**, not just fit.

DiMasi et al., *J Health Econ* 2016; Wong et al., *Biostatistics* 2019.

The mechanistic gap

Statistical / empirical models

- interpolate *within* observed data
- “what” happened — fit to endpoints
- struggle to extrapolate to new doses, populations, combinations

*To predict a drug that does not yet exist, you must model the **mechanism** it will act on.*

Mechanistic (QSP) models

- encode *causal* biology: target → pathway → disease
- “why and how” — simulate the unobserved
- extrapolate across dose · species · subpopulation · regimen

What is Quantitative Systems Pharmacology?

QSP = **systems biology** × **pharmacokinetics/pharmacodynamics**
— a mechanistic, mathematical description of the causal chain

drug → **target** → **pathway** → **disease** → **patient**

It answers questions statistics cannot:

- If we modulate *this* target, how far downstream does the effect propagate?
- Which **patient subtypes** respond — and why?
- What dose, schedule, or **combination** optimizes the efficacy-safety balance?

The anatomy of a QSP model

A QSP model is an initial-value problem — a coupled nonlinear dynamical system:

$$\frac{d\mathbf{x}}{dt} = \mathbf{f}(\mathbf{x}(t), \theta, \mathbf{u}(t)), \quad \mathbf{x}(0) = \mathbf{x}_0$$

States $\mathbf{x}$

drug concentrations ·
receptor occupancy ·
signaling species · cell
populations ·
biomarkers · endpoints

Parameters θ

rate constants · EC_{50} ,
 E_{max} , Hill n ·
clearances · binding
affinities

Inputs $\mathbf{u}(t)$

dosing regimens
(amount, route,
schedule) — the
controllable handles of
therapy

Our library: **15-35+ coupled ODEs** and **70-100+ parameters** per disease.

The recurring nonlinear structures

The same mathematical motifs appear across every disease — a shared grammar:

Hill / E_{max} pharmacodynamics

$$E(C) = \frac{E_{max} C^n}{EC_{50}^n + C^n}$$

Indirect-response (turnover)

$$\frac{dR}{dt} = k_{in} f(\cdot) - k_{out} R$$

Logistic / carrying-capacity growth

Mass-action coupling
(immune-complex formation)

$$\frac{d[IC]}{dt} = k_f [A][B] - k_c [IC]$$

Target-mediated drug disposition (TMDD)

$$\frac{d[D]}{dt} = -k_{on}[D][T] + k_{off}[DT] - \dots$$

Power-law feedback & threshold switching

Why QSP matters — Model-Informed Drug Development

Stage	What QSP contributes	Effect on attrition
Target validation	does target modulation reach the phenotype? bypass / resistance	kills implausible targets early
Mechanism of action	quantifies the causal structure; biomarker strategy	stronger evidence packages
First-in-human dose	exposure-response & safety margins from mechanism	safer, leaner Phase 1
Trial design	virtual patients → enrichment, endpoints, sample size	higher trial success

The bottleneck

A single QSP model traditionally costs **expert-months to years**:

- deep literature synthesis across pathophysiology + pharmacology
- careful ODE formulation and parameterization
- software implementation, calibration, visualization

Consequences

- only a handful of diseases ever get one
- models are siloed, rarely open, rarely reproducible
- the long tail of diseases is left unmodeled

One disease $\approx$ one (or more) expert-years.

Scarce · slow · closed.

Thesis — LLM-augmented model generation

*Use an **autonomous LLM coding agent** to build complete, quality-gated, fully-referenced QSP models — at the pace of one disease per day — with every artifact under version control.*

Each model ships four artifacts

- 🗺️ Graphviz **mechanistic map**
- ⚙️ mrgsolve **ODE model**
- 🖥️ **Shiny** dashboard
- 📖 curated **references**

Built-in principles

- enforced **quality gates**
- **literature grounding** for every parameter
- standardized schema → comparable & auditable
- **reproducible**: code + git history

Architecture

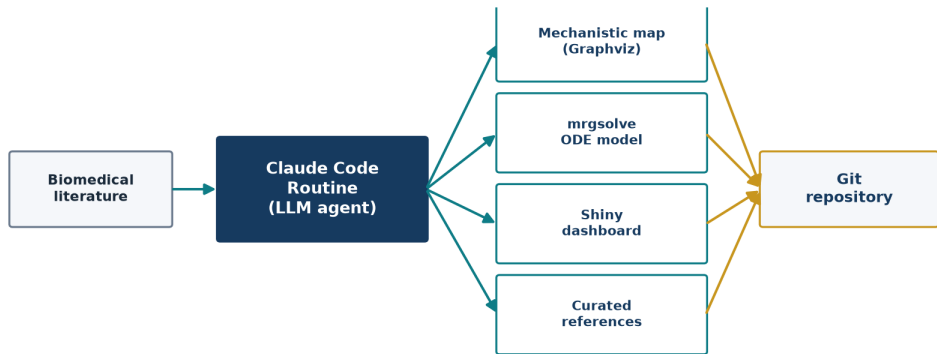
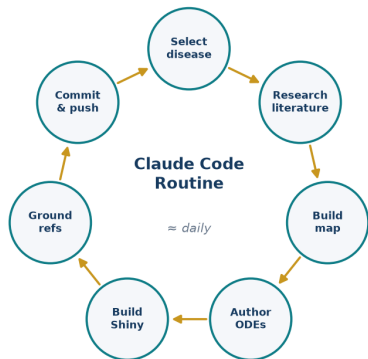


Figure 1: From an autonomous agent to four interoperable artifacts, committed to git.

The Claude Code Routine — an autonomous daily loop ★



A scheduled coding agent. **Each run completes one disease end-to-end and pushes it.**

1. **Select** an uncovered disease (rotate categories)
2. **Research** — synthesize ~50 PubMed sources
3. **Map** — Graphviz network (≥ 100 nodes)
4. **Model** — mrgsolve ODEs (≥ 15 compartments)
5. **Dashboard** — Shiny (≥ 6 tabs)
6. **Ground** — sectioned references (≥ 30)
7. **Commit & push** — automatically

A git **stop-hook** refuses to end a session until everything is committed & pushed —

Inside one session — the guardrails ★

How do we trust code an LLM wrote? **Structure + grounding + verification.**

Quality gates (hard minimums)

- mechanistic map: **≥ 100 nodes, ≥ 8 clusters**
- ODE model: **≥ 15 compartments, ≥ 5 scenarios**
- dashboard: **≥ 6 tabs**
- references: **≥ 30 PubMed-indexed**

Structural templates

- reusable ODE motifs (Hill, turnover, TMDD) shrink the space of “things that can go

Literature grounding

- every parameter set carries a calibration memo citing trials → the principal **anti-hallucination** mechanism

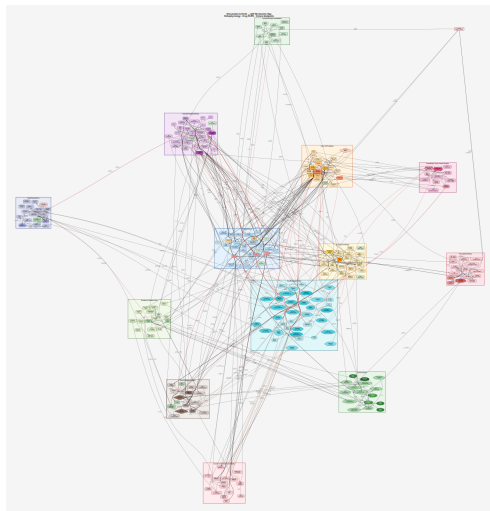
Reproducibility & review

- everything is code in **git**: diffable, auditable, re-runnable
- **human-in-the-loop** scientific review before use

Why an LLM/AI is the right engine ★

- **Tireless & continuous** — runs 24/7, no fatigue; a complete, multi-artifact disease model **nearly every day** — a cadence no human team sustains
 - **Vast literature synthesis** — integrates ~**9667 citations** (~51/model), reading & organizing evidence far faster than manual review
 - **Breadth across diseases & drugs** — 191 diseases and **hundreds of drugs/targets** in one framework
 - **Consistency & standardization** — identical schema and quality bar for every model → comparable, reusable
 - **Speed & economics** — expert-months → **hours**
 - **Democratization** — brings scarce systems-pharmacology expertise to the **long tail** of diseases that would never get a hand-built model
- The scarce resource is modeling expertise. The LLM *scales* it.

Deliverable ① — the mechanistic map (graph theory)



- **Vertices** = species / states;
edges = mechanistic interactions (activation, inhibition, mass transfer)
- **Clusters** = pathway modules — a community structure over the graph
- Built with **Graphviz**; rendered to SVG / PNG and version-controlled
~11.7 **pathway modules per model**
— up to 685 graph elements each.

Deliverable ② — the mrgsolve ODE system

Real excerpt — IgA nephropathy

“four-hit” cascade:

```
// Hit 3: immune-complex formation
```

```
dxdt_IC_mes = k_form*GdIgA1*AutoAb - k_deg*IC_mes;
```

```
// complement amplification (iptacopan)
```

```
dxdt_CompAP = k_syn*IC_mes*(1-E_ipt) - k_deg*CompAP;
```

```
// Hill drug effect; Bliss-independent
```

```
E_bud = Emax*pow(C,h)/(pow(EC50,h)+pow(C,h));
```

```
E_muc = 1-(1-E_bud)*(1-E_sib);
```

- compiled **C++** solver (stiff systems, event-based dosing)

• **coupled drug PK → disease**

• **PD-leads to clinical endpoints**

• **15-35k** state variables per model

• **≥ 5 therapeutic scenarios** each

• **Genuinely** runnable systems

• pharmacology — not pseudocode.

Deliverables ③ + ④ — interaction & evidence

③ **Shiny dashboard** — interrogate the model without code

- patient profile · PK · pathophysiology
- clinical endpoints · scenario comparison · biomarkers (6–8 tabs)

Shiny dashboard

[screenshot placeholder — drop a capture here]

④ **Reference grounding** — traceable evidence

- ~**51** PubMed citations per model, sectioned by mechanism
- every parameter & claim anchored to the literature
- ~**9667** references across the library

Grounding is what separates a model from a guess.

A living library, growing daily

Metric	Value
Disease models	191 ($\approx +1$ / day)
Therapeutic areas	~ 15
mrgsolve ODE systems	192
Pathway clusters (graph modules)	~ 2228
PubMed references	~ 9667

Each is a complete, runnable, fully-referenced QSP model.

The gallery wall

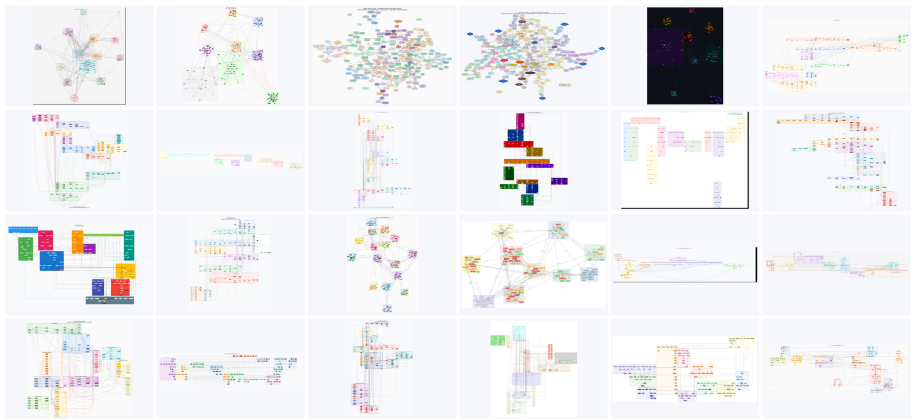


Figure 3: A fraction of the 191 mechanistic maps — one autonomous build each.

Breadth of coverage

One consistent framework, spanning the spectrum of human disease:

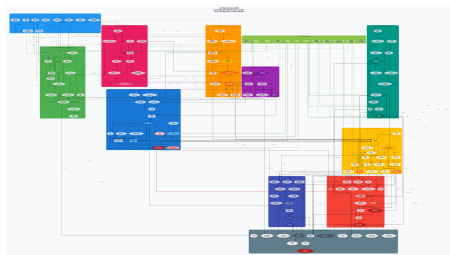
Oncology — solid · Oncology — heme · Autoimmune / rheumatic ·
Vasculitis · Cardiovascular · Respiratory · Renal / urologic · GI /
hepatobiliary · Endocrine / metabolic · Neurologic · Psychiatric ·
Dermatologic · Infectious · Ophthalmic · Rare / genetic

- **Cancers** — breast, NSCLC, SCLC, glioblastoma, CML, multiple myeloma, melanoma, pancreatic...
- **Rare & genetic** — Fabry, Gaucher, DMD, SMA, Huntington, transthyretin amyloidosis...

- **Common chronic** — diabetes, heart failure, COPD, CKD...
- **Immune / hematologic** — lupus, RA, sickle cell, ITP, myelofibrosis...

Hundreds of distinct **drugs and molecular targets**.

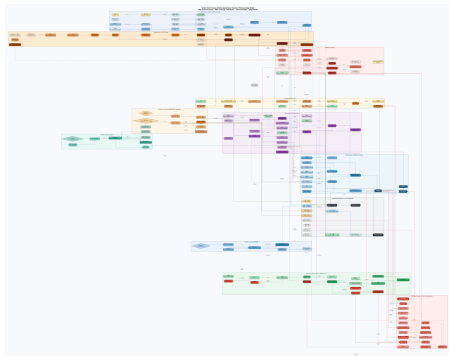
IgA nephropathy — the “four-hit” cascade



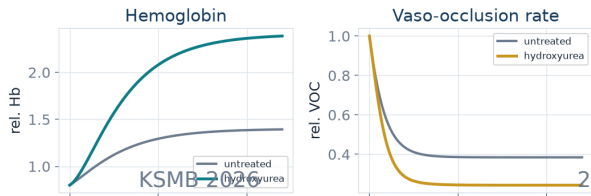
- **20 ODEs:** Gd-IgA1 → autoantibody → immune complex → complement → mesangial / podocyte injury → UPCR, eGFR
- mass-action coupling + Hill drug effects + **Bliss-independent** combinations
- **7 scenarios:** RAAS, budesonide-TRF, sparsentan, iptacopan, sibeprenlimab, triple therapy
- calibrated to **NeflgArd, PROTECT, APPLAUSE-IgAN**



Sickle cell disease — polymerization to vaso-occlusion



- **24 ODEs:** HbS polymerization → hemolysis → **NO scavenging** (bimolecular) → P-selectin → vaso-occlusion
- **power-law feedback:** erythropoiesis $\propto (Hb_0/Hb)^\gamma$
- drugs: hydroxyurea (HbF induction, Hill $n=1.5$), voxelotor, crizanlizumab, L-glutamine
- calibrated to **MSH, HOPE, SUSTAIN**



The mathematics of the *whole* library

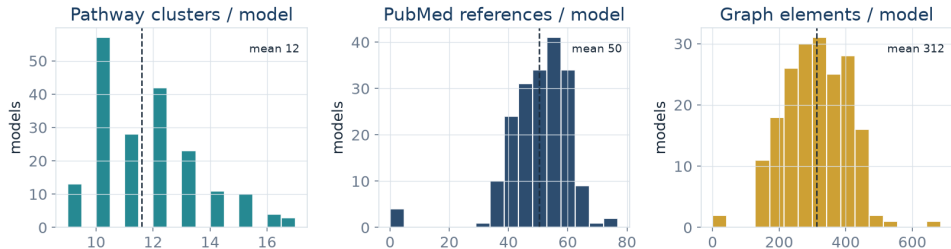


Figure 7: Distributions across 191 models: pathway clusters, references, and graph size.

A standardized corpus of nonlinear dynamical systems — itself an object worth studying.

Open mathematical problems — a collaboration invitation

A corpus of ~191 nonlinear ODE systems is a playground for applied mathematics:

- **Identifiability** — structural & practical: which θ are recoverable from feasible measurements?
- **Sensitivity analysis** — global (Sobol, Morris) ranking of drivers
- **Model reduction** — timescale separation, QSSA, balanced truncation of {15-35}-D systems
- **Uncertainty quantification** — parameter & structural uncertainty → prediction intervals
- **Dynamics** — stability, bifurcations, thresholds (adaptive → maladaptive RV failure in PAH)
- **Inverse problems & virtual populations** — calibrate parameter *distributions* to data

191 models, one schema → methods you develop transfer across all of them.

Validation & limitations — stated plainly

What these models are

- mechanistically structured, literature-grounded
- calibrated to **published trial endpoints**
- excellent for hypothesis generation, teaching, MIDD scaffolding

*Honesty about scope is part of the method. The LLM drafts; **science validates.***

What they are *not* (yet)

- **not** fitted to patient-level data
- **not** for clinical decision-making
- a plausible-looking model can still be wrong → references + human verification are essential; formal validation is future work

Reproducibility & open science

- Every model = **code + map + references**, openly in one git repository
- Each is independently **runnable** (mrgsolve) and **explorable** (Shiny)
- Full **history**: who / what / when for every parameter — diffable and auditable
- The library **grows in public**, one verifiable commit at a time

100% open · versioned ·
reproducible

Traditional vs. LLM-augmented QSP

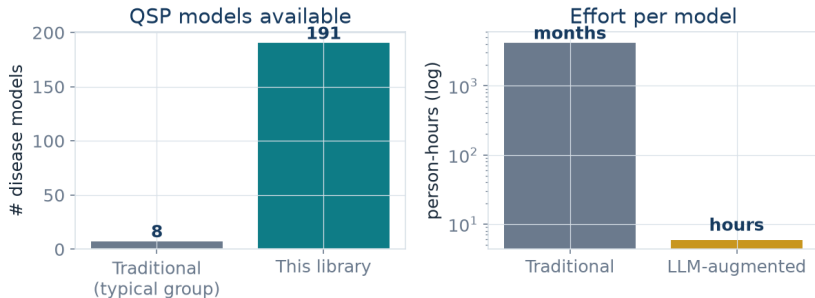


Figure 8: Coverage and per-model effort: a step change, not an increment.

Future directions

- **Patient-level calibration** — Bayesian fitting to real PK / PD & EHR data
- **Virtual populations** — sample parameter distributions → trial simulation
- **Automated model reduction** — derive minimal mechanistic cores
- **Coupling** to clinical pharmacometric pipelines and electronic health data
- **Formal verification** of generated ODE structure & units
- **Community contributions** — an open, extensible standard for mechanistic disease models

Conclusion

- Drug development needs **mechanism**, expressed as **nonlinear ODE systems**
- QSP supplies it — but has been **slow, scarce, and closed**
- An **autonomous LLM agent** changes the economics: **191** complete, grounded, reproducible models — **and one more nearly every day**
- The result is a **standardized mathematical corpus** of human disease — open to *your* methods

Democratizing mechanistic modeling — at the scale only tireless AI makes possible.

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An LLM-Augmented QSP Platform



With thanks to all co-authors and to PIPET & TiumBio.

Thank you. Questions welcome.